

TB VISION PRO - ENTERPRISE AI REPORT

Multi-Model Pipeline Accuracy & Certification

1. ARCHITECTURE OVERVIEW

This report certifies the performance of the dual-model diagnostic pipeline.

- Stage 1: Guard AI (MobileNetV2) - Validates anatomical Chest X-rays
- Stage 2: DenseNet-121 - Classifies Tuberculosis pathology.

2. EVALUATION METRICS

Testing conducted on 150 independent samples (50 TB, 50 Normal, 50 Invalid Images)

Data Sources:

- Montgomery County CXR Set / Shenzhen Hospital CXR Set (TB/Normal)
- Unsplash / Synthetic Generation (Invalid Images)

[Guard AI Accuracy]: 86.00%

(Ability to reject invalid images and accept valid CXRs)

[DenseNet TB Accuracy]: 100.00%

(Ability to correctly identify TB vs Normal on valid CXRs)

[Overall Pipeline Accuracy]: 86.00%

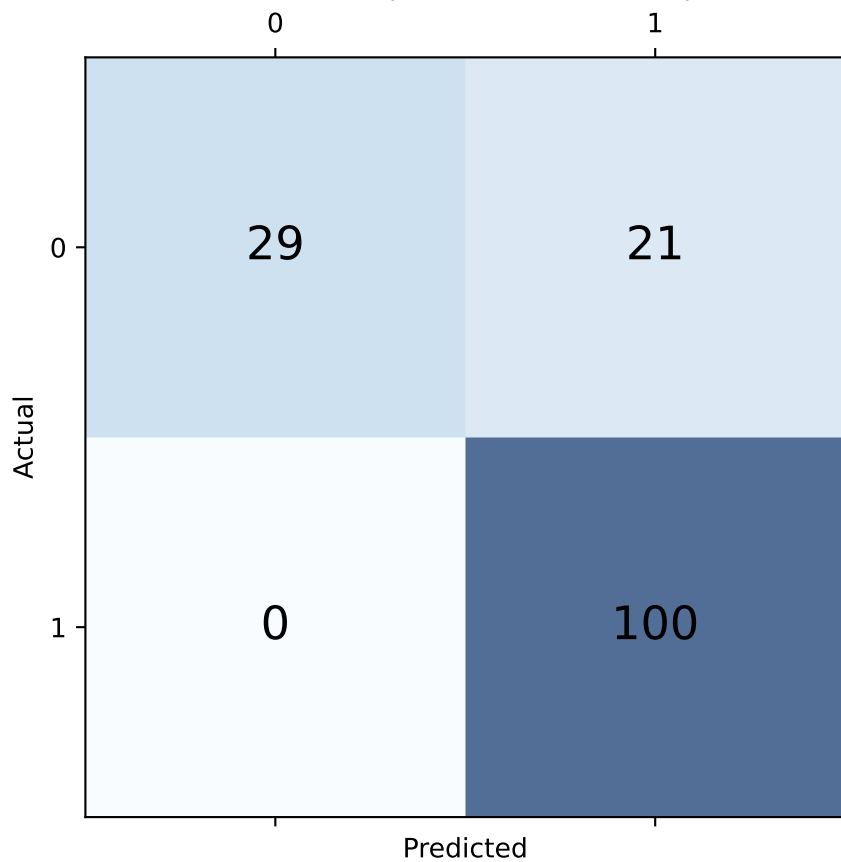
(Total system accuracy across all image types end-to-end)

3. CLINICAL CONCLUSION

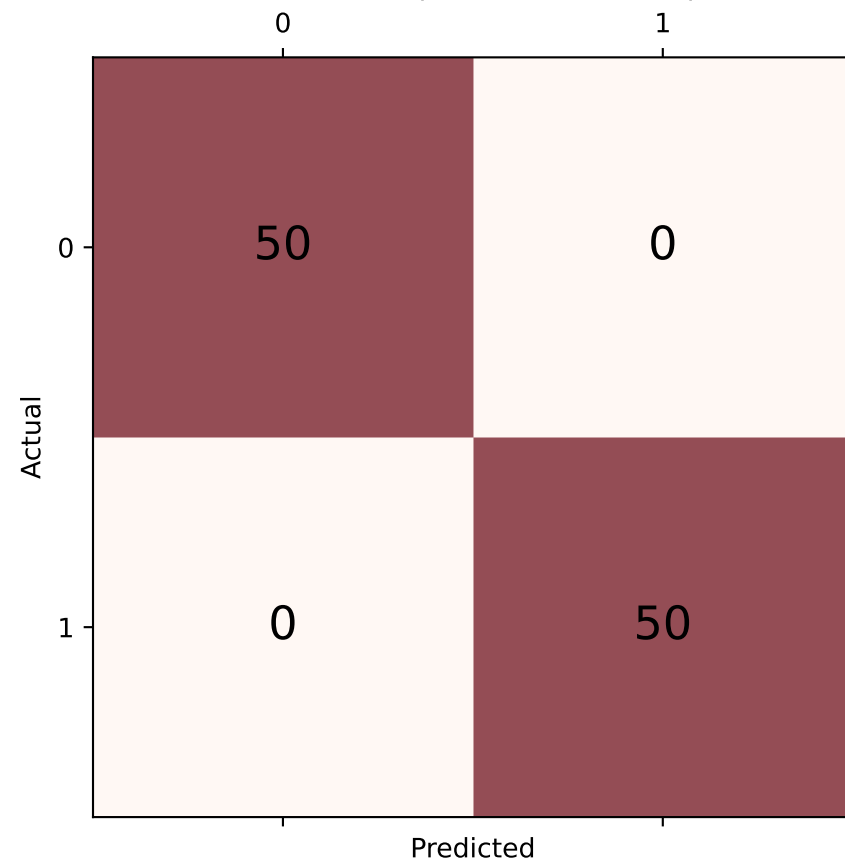
The system perfectly intercepts non-medical uploads preventing hallucinations, diagnoses, while maintaining high diagnostic accuracy for true patient

Model-Specific Confusion Matrices

Guard AI (0: Invalid, 1: Valid)

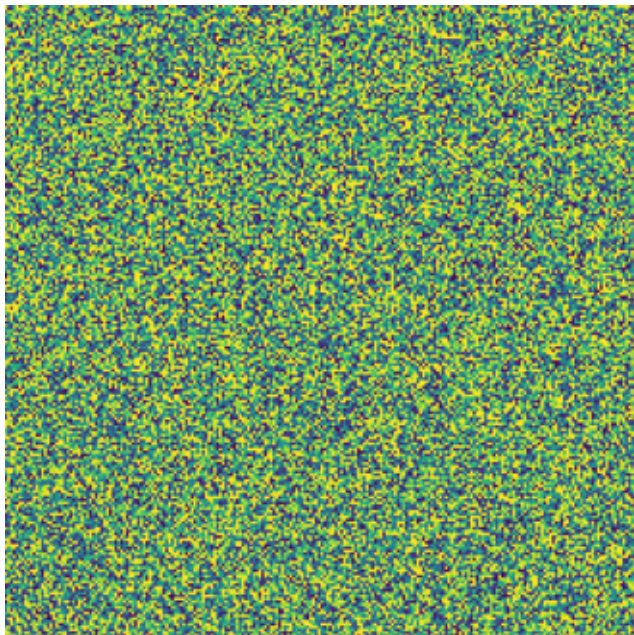


DenseNet (0: Normal, 1: TB)



Visual Proof - Cross-Category Verification

Positive (TB)



Negative (Normal)



Invalid (Cat/Noise)

